Saurav

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Profile

Robotics & autonomous systems researcher specializing in reinforcement learning, physics-informed machine learning, and UAV/ground-robot simulation. Experience spans PPO-based teleoperation policies, physics-informed generative models for structural health monitoring, and multi-body dynamics validation for field robots – backed by an incoming research position at IISc Bangalore. Comfortable operating across the full stack, from ROS2/Gazebo simulation to FEA-validated hardware design.

Education

Indian Institute of Information Technology, Design & Manufacturing, Kurnool 2021 – 2025
B.Tech Mechanical Engineering (Specialization: Design & Manufacturing) *70% (3.0/4.0)*

Certifications

- ROS2 (Jazzy), TF, URDF, RViz, Gazebo, Isaac Sim & Nav2 with SLAM

Honors & Awards

International SciML + GenAI Challenge | **Vizuara AI Labs (Conducted by Raj Dandekar, PhD, MIT)** | Code | certificate Mar 2026

- **Runner-up (2nd of 10 finalist teams).** Architected a generative-AI framework for synthesizing physically consistent vibration signatures for structural damage modes in drone wings, addressing the data-scarcity problem inherent to rare real-world failure events.
- Formulated a parametric Physics-Informed Neural Network (PINN) embedding the Euler–Bernoulli beam equation as a hard constraint in the training loss, ensuring every synthetic sensor time-history satisfies conservation of energy and momentum by construction.
- Synthesized a corpus of **100,000+ physics-verified, labeled acceleration time-histories** spanning a continuous damage parameter space (location and severity), enabling lightweight onboard classifiers to generalize to unseen failure modes with >**95% detection accuracy**.

Research Experience

Research Intern (Incoming) – Robotics Innovations Lab, IISc Bangalore Aug 2026
Supervisor: Prof. Abhra Roy Chowdhury

- Developing RL-based teleoperated robot navigation strategies for complex, unstructured environments, combining reinforcement learning with human-in-the-loop teleoperation for adaptive, robust control.
- Designing a transformable climbing robot for pipeline inspection, focused on passive diameter adaptation, multi-body dynamics, and structural validation using PyBullet and FreeCAD/CalculiX.

Mech-Team Lead, Research Project (funded by IIT Hyderabad) March 2023 – Dec 2024
UAV Calibration & Flight Testing *Supervisor: Dr. Eswaramoorthy KV*

- Led a 4-member team integrating, calibrating, and validating Pixhawk/ArduPilot-based autonomous drone platforms through systematic flight testing.
- Developed standardized calibration and deployment protocols, reducing pre-flight setup time by **30%** and enabling reproducible operations across multiple UAV configurations.

Undergraduate Research Intern, NIT Patna | certificate Dec 2024 – Mar 2025
Composite Mechanics & Blast-Response Modelling

- Developed MATLAB-based finite-element models to study the transient blast response of jute-fiber-reinforced composite structures.
- Analyzed deformation and energy-absorption behavior under blast load, achieving **12%** agreement with published experimental benchmarks.

Undergraduate Research Intern, IIT (ISM) Dhanbad | certificate May 2023 – Jul 2023
Micromachining and Automation

- Designed a closed-loop RPM measurement and control module for a semi-high-speed micromachining center using Arduino Uno, embedded within the Magnetic Levitation EDM (Maglev EDM) research platform.
- Integrated the RPM feedback loop into the spindle control architecture with doctoral researchers, yielding a **20% improvement in rotational speed stability** and enhancing surface-finish repeatability.
- Executed end-to-end additive manufacturing of a high-fidelity freeform geometry, including toolpath generation, support-structure design, and surface-finish optimization.

Open Source Contributions

Dora Middleware (v0.4.1) | Pull Request 2026

- Upgraded the LLM tutorial infrastructure by implementing class-based operator patterns, improving code maintainability and pedagogical clarity for new users.
- Enhanced error handling and logging across the tutorial pipeline; passed all **27 CI checks** and received maintainer approval.

Publications & Conferences

ICoIED 2026 (IIT Roorkee) | certificate 2026

- Engineered an AA6061/SiC (5 wt.%) metal-matrix composite via stir casting and executed a Taguchi L16 design-of-experiments campaign for CNC turning across 16 experimental runs, constructing a multivariate machinability response surface.
- Identified Pareto-optimal process parameters minimizing surface roughness and tool wear while maximizing material-removal rate, using a hybrid MEREC–MAIRCA multi-criteria decision-making framework; presented at ICoIED 2026, IIT Roorkee.

Robotics & Autonomous Systems Projects

Shared-Autonomy Framework for Human-Aware Teleoperated Inspection | *RL, PPO, Gazebo, ROS2* Jan – Apr 2026

- Formulated a teleop-conditioned PPO policy with a 26-dimensional observation vector encoding the operator's live velocity command alongside 24 downsampled LiDAR range beams.
- Derived a continuous three-tier authority-blending law (full human authority → graduated assistance band → reactive backstop) driven by real-time LiDAR proximity, eliminating discontinuities from discrete mode-switching.
- Trained the PPO agent for 50,000 steps in a ROS 2 Jazzy / Gazebo Harmonic warehouse world with dynamic human agents; achieved **93% human-avoidance rate** and **91% task-completion rate**, outperforming pure-teleoperation and full-autonomy baselines.

Generative Design & Structural Validation of a Transformable In-Pipe Inspection Robot | Code Feb – May 2026

- Developed a fully parametric OpenSCAD model of a three-arm radial climbing robot with passive diameter adaptation across 150–300 mm, deriving a closed-form kinematic mapping for arm deployment.
- Built a multi-body dynamic simulation in PyBullet to quantify hinge reaction forces, then validated loads via a second-order tetrahedral FE model in FreeCAD/CalculiX, confirming a **peak von Mises stress of 78.4 MPa** and **safety factor of 3.06** in the Al 6061-T6 arm.
- Established a reproducible parametric CAD → multi-body dynamics → FEA validation pipeline; identified topology-optimization potential to reduce arm mass by 25–35% while preserving safety margins.

Cyber-Physical Systems & UAV Simulation Architecture | Code Mar – Jun 2026

- Designed a cyber-physical simulation architecture integrating ROS2, Gazebo, RViz, and ArduPilot SITL, enabling zero-cost hardware prototyping and pre-flight validation of autonomous flight stacks.
- Engineered modular ROS2 Python control nodes and DroneKit/MAVLink wrappers, achieving sub-10 ms command latency and **precision landing with <0.2 m touchdown error** across 50+ simulated missions.
- Implemented a hierarchical motion-planning stack (global A* + local Dynamic Window Approach), improving obstacle-avoidance success rate by **95%** and reducing path traversal cost by **30%**.

Autonomous Drone Swarm Simulation (Python) | Code Apr – Jun 2026

- Developed a distributed multi-agent swarm architecture with Raft-lite leader election, probabilistic gossip task dissemination, and spatial load balancing, achieving robust consensus under **60% simulated packet loss**.
- Validated fault-tolerance via discrete-event simulation: **sub-3 s leader failover**, autonomous crash recovery, and heterogeneous agent coordination, instrumented with a real-time terminal radar visualization.

Technical Skills

Programming: C++, Python, XML, MATLAB

Robotics & UAV: ROS2, URDF, Gazebo, RViz, Isaac Sim, Nav2 with SLAM, Pixhawk, ArduPilot, DroneKit, MAVLink

Simulation & Tools: PyBullet, FreeCAD/CalculiX, OpenSCAD, Stable-Baselines3, PyTorch

Research: MCDM (TOPSIS, MEREC), Simulation, System Modeling, Linux (Ubuntu), Git, LaTeX

Activities

- **XPRENEURS Startup Talent Hour (UnternehmerTUM)** — Selected from a highly competitive international applicant pool; engaged with 20 deep-technology ventures across the Munich startup ecosystem and secured internship offers from Munich-based companies.